ASAF REUCHER CORREA ARROYO

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RESEARCH INTERESTS

Low-resource NLP, machine translation, and morphology-aware tokenization; efficient model compression; applied AI for urban safety and social impact; multimodal grounding; sample-efficient language learning.

PUBLICATIONS

A. R. Correa Arroyo and R. Correa Morocho, “Fine-Tuning a 600M NLLB-200 Model for Low-Resource Spanish-Quechua Machine Translation,” Accepted for Oral Presentation at INTERCON 2026 (IEEE). *Code available upon request.*

RESEARCH EXPERIENCE

Student Research Assistant | *Media & Information Interaction Group, Idiap Research Institute* (Remote)
August 2026 - present

Project: A Cross-Cultural Study of Podcasts for Well-Being Information Dissemination

- Conducted literature review on topic modeling and evaluation methods.
- Performed data preparation, processing, and data engineering for a multilingual podcast corpus.
- Designed and executed LLM-based experiments for content analysis.
- Skills: NLP, topic modeling, multilinguality.

Independent Researcher ***2026 – Present***

Runasimi-MT: Neural Machine Translation for Agglutinative Low-Resource Languages (Quechua, Aymara)

- Fine-tuned NLLB-200-distilled-600M on Spanish-Quechua/Aymara corpora (chrF++ 42.68 and 43.10); see Publications.

*Morphology-Aware Tokenization for Agglutinative NMT — **In Progress***

- Designing a Quechua/Aymara-specific subword tokenizer to address suffix-boundary fragmentation limiting the Runasimi-MT results. Code available upon request.

UNDERGRADUATE RESEARCH (FACULTY-SUPERVISED)

Secure Gait Recognition System — Dr. Wonjun Lee ***Sep 2025 – Dec 2025***

- Collaborated in a team of three to develop a biometric security system using Spatial-Temporal Graph Convolutional Networks (ST-GCN), analyzing adversarial resilience against spoofing and replay attacks via MediaPipe skeletal data.

NLP for Low-Resource Dialects — Dr. Yohan Lee, Dean ***Jan 2025 – Dec 2025***

- Designed matrix encodings for bilingual text pairs targeting Peruvian Spanish medical dialects; validated translation outputs in TensorFlow, addressing colloquialism and emotional-accuracy challenges in healthcare interpretation.

LLMs for Stock-Keeping-Unit Generation — Dr. Shieu-Hong Lin ***Jan 2025 – May 2025***

- Fine-tuned Transformer models for inventory description generation; designed exact-match and edit-distance evaluation protocols, optimizing for low-resource inference environments.

APPLIED ML ENGINEERING PROJECTS

Cascade System for Context-Aware Urban Violence Detection ***Jul 2026***

- Developed a three-stage cascade architecture for urban surveillance video: person tracking (YOLOv8 + ByteTrack), weapon detection (YOLOv9s on SOHAS), and temporal classification (TriPath3DNet on R3D-18).

- Designed a bounded additive boost mechanism for contextual support variables (weapon presence, proximity) calibrated via constrained search, avoiding overfitting observed in a prior higher-freedom formulation.
- Built with PyTorch and Google Colab, with a focus on computational feasibility for edge deployment.

Network Intrusion Detection with Hybrid CNN-LSTM and Autoencoder ***Aug 2026***

- Designed a multi-class intrusion detection system on CICIoT23 (5.5M training rows) combining an LSTM autoencoder for unsupervised anomaly detection and a hybrid CNN-LSTM classifier for 34 attack classes.
- Implemented streaming feature scaling and chunk-based data pipelines to handle large-scale datasets within memory constraints.
- Explored sparsity techniques on dense layers and identified the need for structural compaction to achieve actual model size reduction, using TensorFlow/Keras and Google Colab.

EDUCATION

Biola University — La Mirada, CA, USA ***Aug 2023 – Dec 2025***

B.S. in Computer Science: Data Science & Information Systems; Minor in Biblical and Theological Studies

- GPA: 3.617 / 4.00 (Cum Laude). Completed a four-year curriculum in 2.5 years; graduated at age 19.
- Honors: President's Scholarship, Dean's List (Sep 2024 – Dec 2025), Larry Acosta Scholarship, Good Shepherd Scholarship.
- Relevant coursework: Deep Learning, Theory of Algorithms, Data Structures, Nature of Computing, Probability & Statistics, Advanced Topics in Mathematics.
- William Lowell Putnam Mathematical Competition — participated 2023, 2024, 2025; scored 3 points in first-time participation (2023), recognized in university news.

TECHNICAL SKILLS

Python, PyTorch, TensorFlow, Keras, Hugging Face Transformers, C++, SQL, Git, Linux, LaTeX. Spanish, English.

LEADERSHIP & TEACHING EXPERIENCE

Cybersecurity Club — Founder & President, Biola University ***Jan 2025 – Dec 2025***

- Joined as an NCL participant (Fall 2023); led the team (Spring 2025) and formalized the club (Fall 2025); organized weekly workshops and established a mentorship pipeline.

Teaching Assistant — Biola University ***Jan 2025 – Dec 2025***

- TA for Nature of Computing, Data Structures, and Programming Languages; mentored students on algorithms and debugging, and assisted in grading technical assignments.